

PRACTICAL COURSE WORKBOOK

Cloud Security - AWS & GCP

Apply practical protection to accounts and systems

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a prioritised security review and action plan
SKILLS	AWS, GCP, IAM, Zero Trust, Critical evaluation, Documentation
REVIEWED	2026-08-25

WHAT YOU WILL BE ABLE TO DO

- Use AWS appropriately in a realistic, bounded task.
- Use GCP appropriately in a realistic, bounded task.
- Use IAM appropriately in a realistic, bounded task.
- Use Zero Trust appropriately in a realistic, bounded task.

WHO THIS IS FOR

Defenders and developers learning lawful, defensive security practice.

BEFORE YOU BEGIN

- Basic networking and operating-system knowledge

Use this guide beside the interactive lessons. Keep inputs, decisions, tests, limitations and the next improvement.

COURSE MAP

From foundation to finished work

Cloud Security - AWS & GCP is a structured, practical course covering AWS, GCP, IAM, Zero Trust. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Security foundations	1. Assets with AWS 2. Threats with GCP 3. Risk with IAM
02 Defensive controls	1. Identity with Zero Trust 2. Network protection with AWS 3. Hardening with GCP
03 Assessment	1. Safe testing with IAM 2. Evidence with Zero Trust 3. Remediation with AWS
04 Operations	1. Monitoring with GCP 2. Response with IAM 3. Recovery with Zero Trust

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Cloud Security - AWS & GCP project using AWS, GCP, IAM.

DELIVERABLE	A working Cloud Security - AWS & GCP artefact plus an evidence-based self-review.
TOOLS	A suitable AWS environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using AWS.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

Use security techniques only on systems you own or are explicitly authorised to test.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Cybersecurity Framework 2.0

NIST - reviewed 2026-08-21
<https://www.nist.gov/cyberframework>

Web Security Testing Guide

OWASP Foundation - reviewed 2026-08-21
<https://owasp.org/www-project-web-security-testing-guide/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the AWS scope.